



## **CLOUD COMPUTING LAB**

**LAB NUMBER 1:**

**LAB TITLE:**

GitHub identity, student portal registration, and ubuntu server installation

**NAME:**

Zahra Akram.

**REGISTRATION NUMBER:**

FA24B1-SE-040

**COURSE CODE:**

SE-665

**SECTION:**

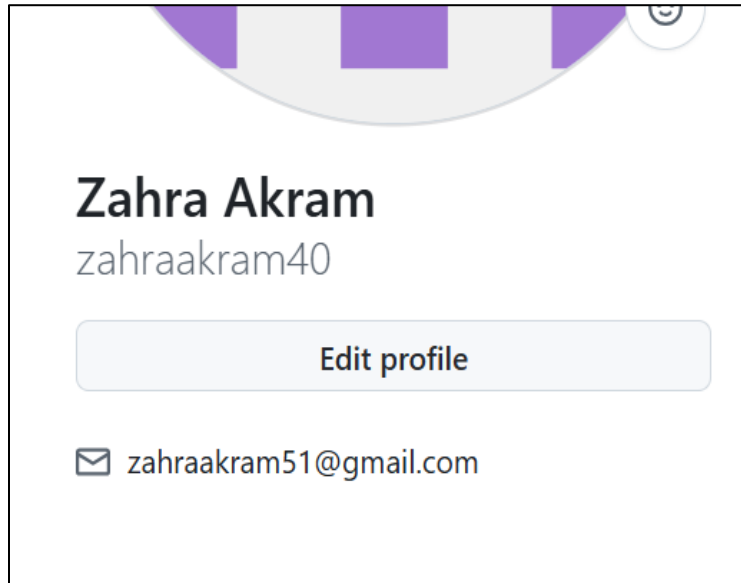
B

**GITHUB USERNAME:**

zahraakram40.

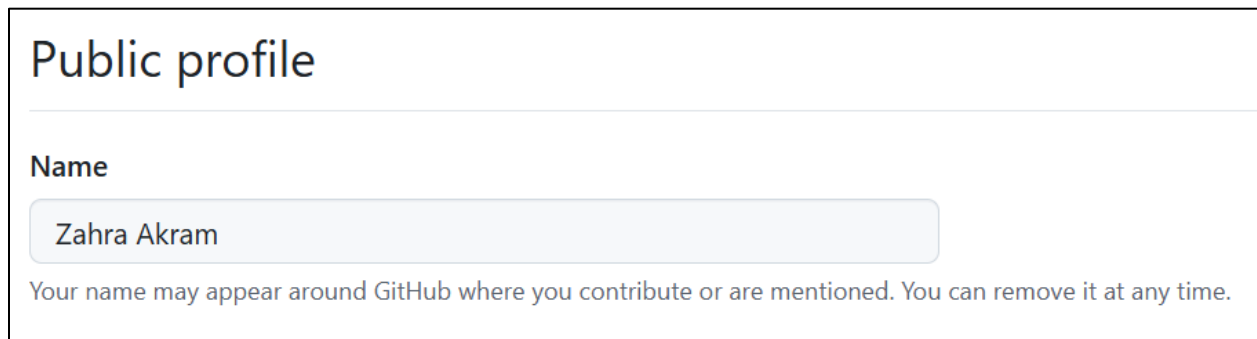
## Task 1: Create and Configure Your GitHub Account

1.github\_profile.png



showing my github profile details.

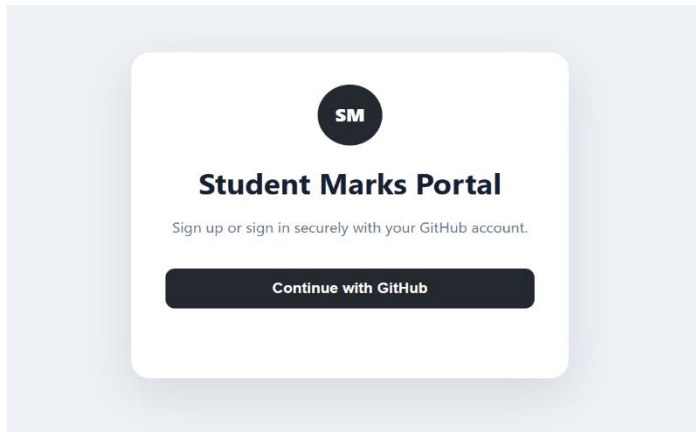
2. github\_actual\_name.png



Showing my actual name in github profile.

## Task 2: Register in the Student Marks Portal

3.portal\_login\_page.png



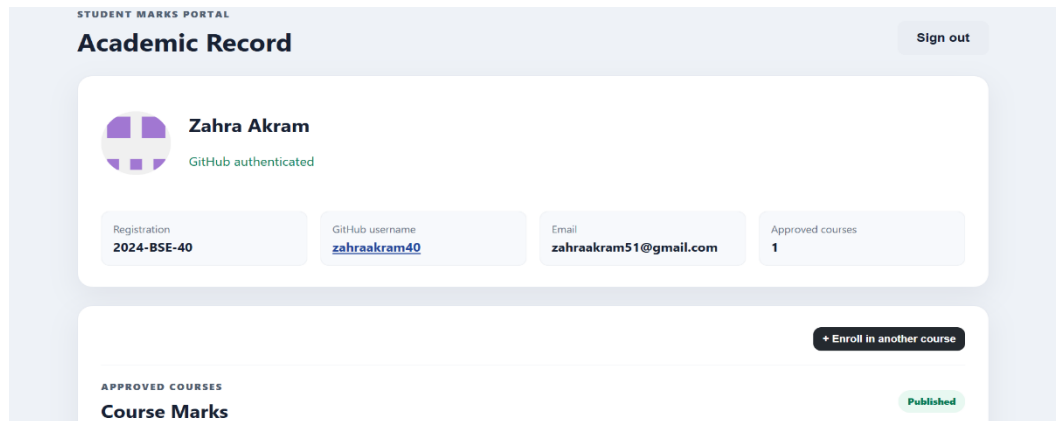
Showing student portal through github

4. portal\_github\_profile.png

A screenshot of the 'Complete registration' form within the portal. At the top center is a dark circular logo with the white text 'ID'. Below the logo, the title 'Complete registration' is displayed in a bold, dark font. Underneath the title, two lines of text read 'Enter your registration number, course and section.' and 'The enrollment requires approval.' The form contains three input fields: 'Registration number' with the value '2024-BSE-40', 'Course' with the value 'SE-665 — Cloud Computing', and 'Section' with the value 'Section B'. Below these fields is a dark rectangular button labeled 'Submit enrollment'. At the bottom of the form is a light gray rectangular button labeled 'Sign out'. The form is framed by a light gray border.

showing your GitHub profile information inside the portal

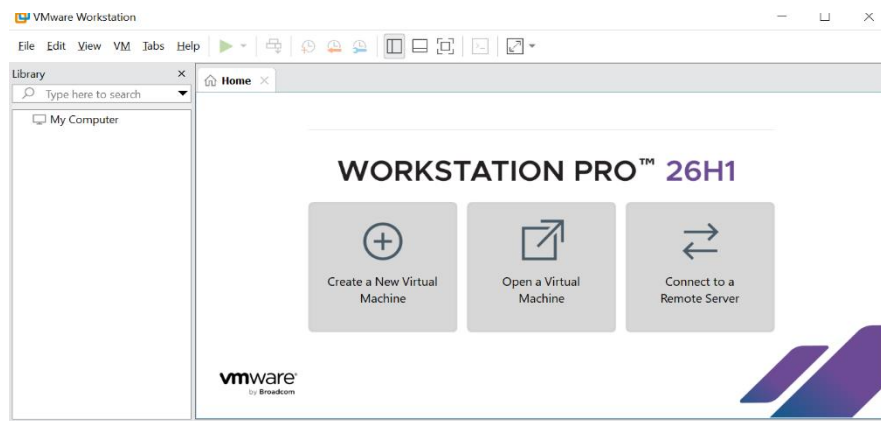
5. portal\_enrollment\_submitted.png



Showing submitted enrollment in student portal.

### Task 3: Download VMware Workstation Pro and Ubuntu Server

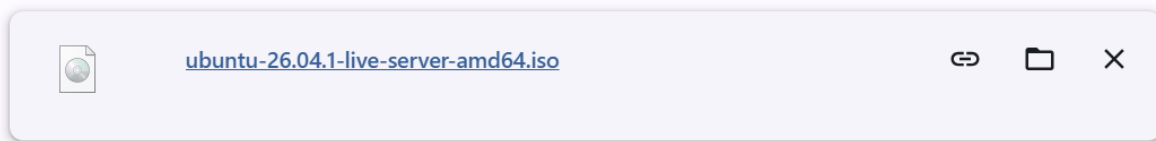
6. vmware\_workstation.png



Vmware workstation pro running successfully.

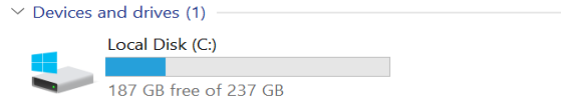
7. ubuntu\_server\_iso.png

September 16, 2026



Showing downloaded ubuntu server iso file.

8. available\_storage.png



Showing available storage on my laptop.

#### Task 4: Create the Ubuntu Server Virtual Machine

9. vm\_creation\_wizard.png



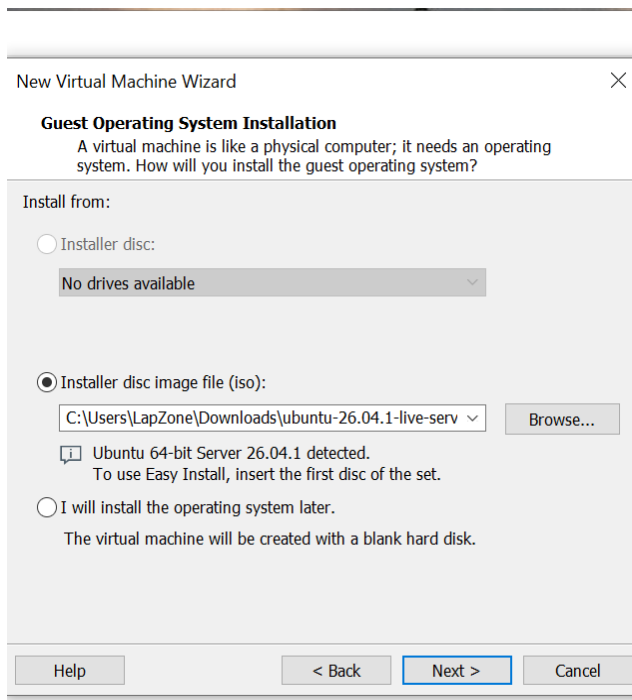
Showing virtual machine wizard.

10. vm\_typical\_configuration.png



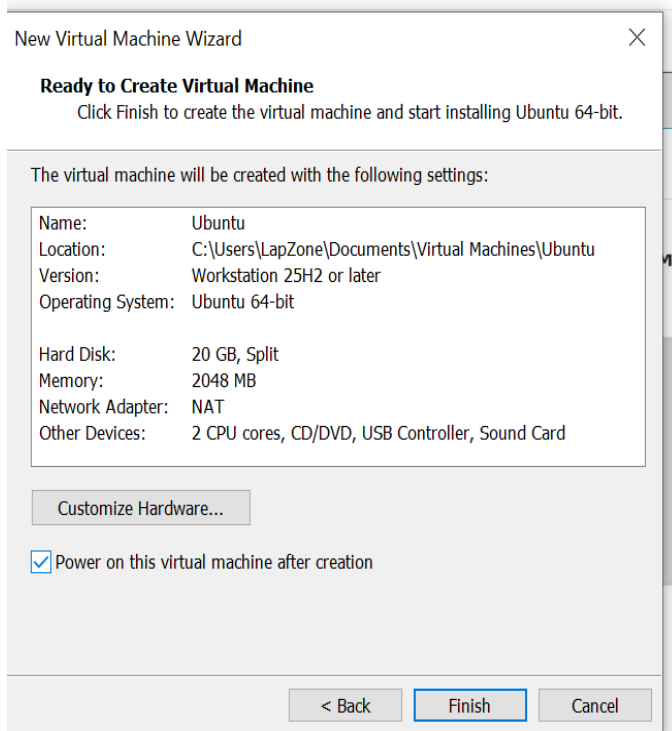
For creating new virtual machine typical option is selected.

11. vm\_ubuntu\_iso\_selected.png



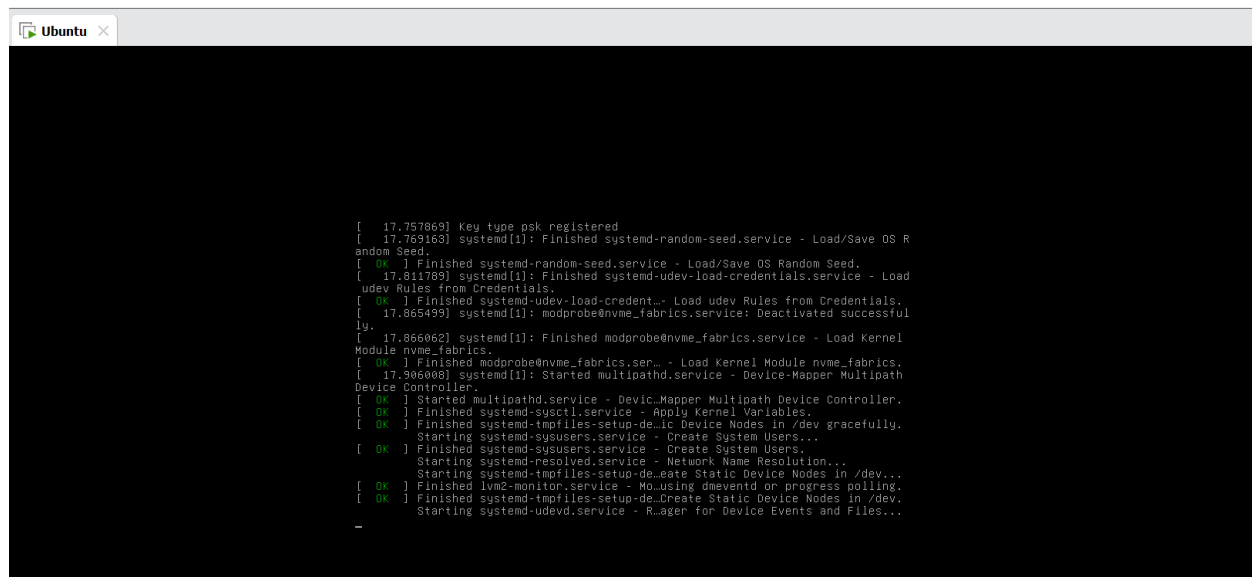
**selected the ubuntu iso path.**

12. vm\_configuration\_summary.png



**showing summary of virtual machine configuration.**

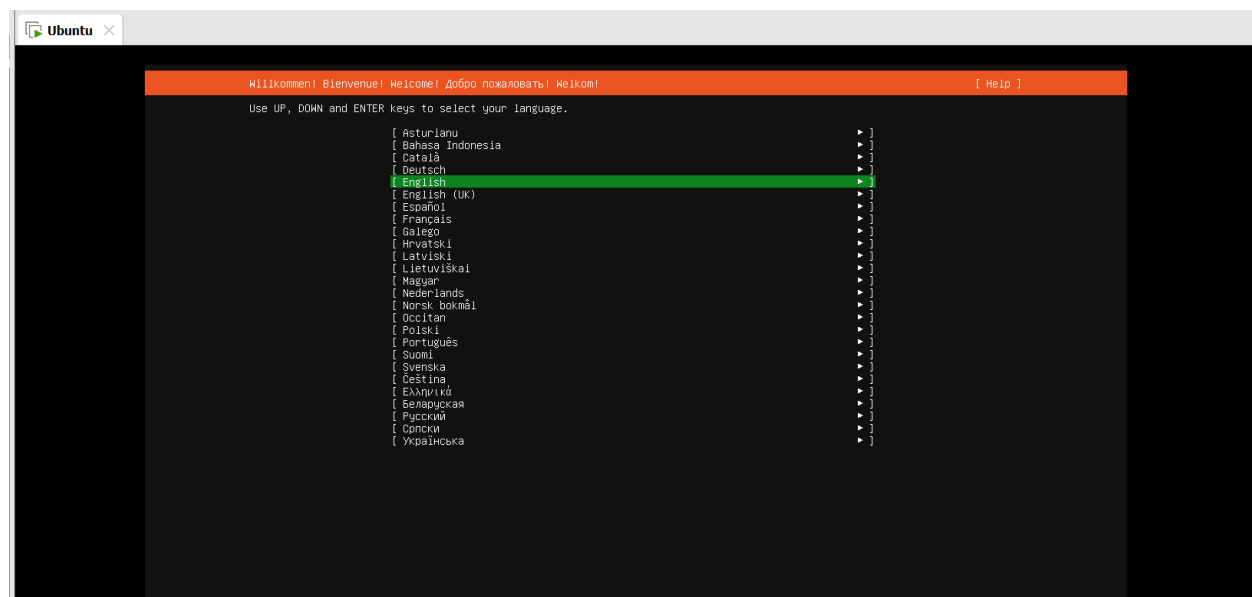
### 13. ubuntu\_installer\_boot.png



Showing ubuntu server installer boot screen.

### Task 5: Choose the Installation Language

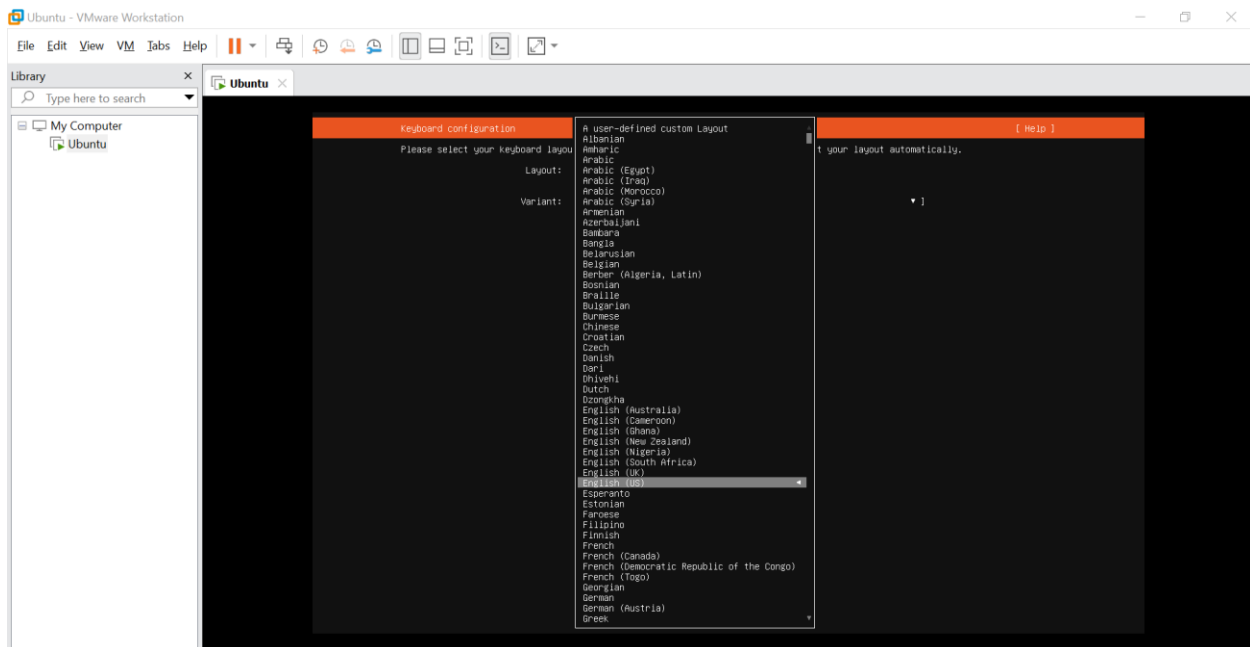
### 14. ubuntu\_language.png



Showing selected ubuntu server language.

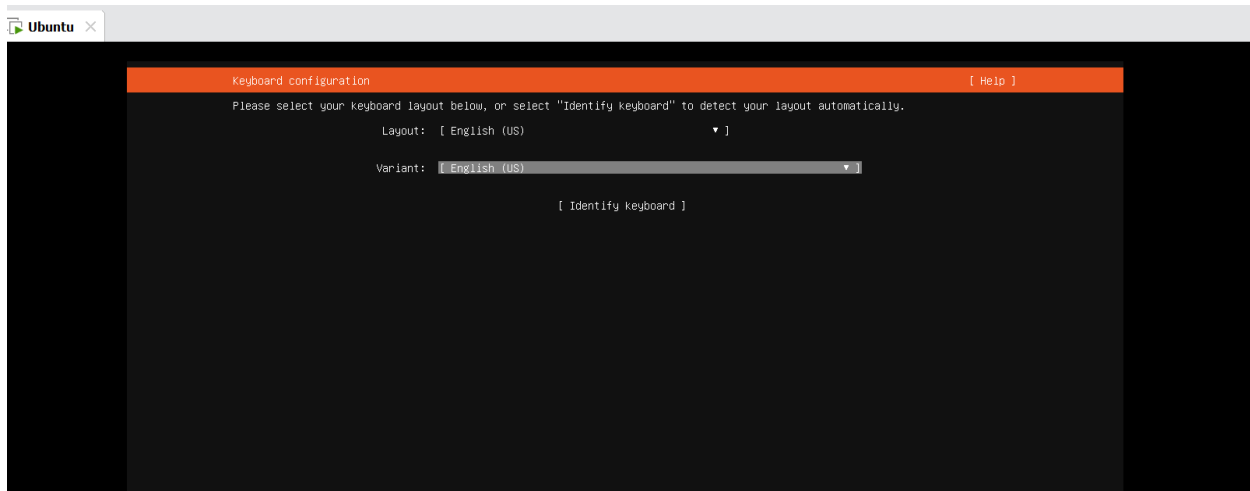
### Task 6: Configure the Keyboard Layout

### 15. ubuntu\_keyboard\_layout.png



Showing selected ubuntu server keyboard layout.

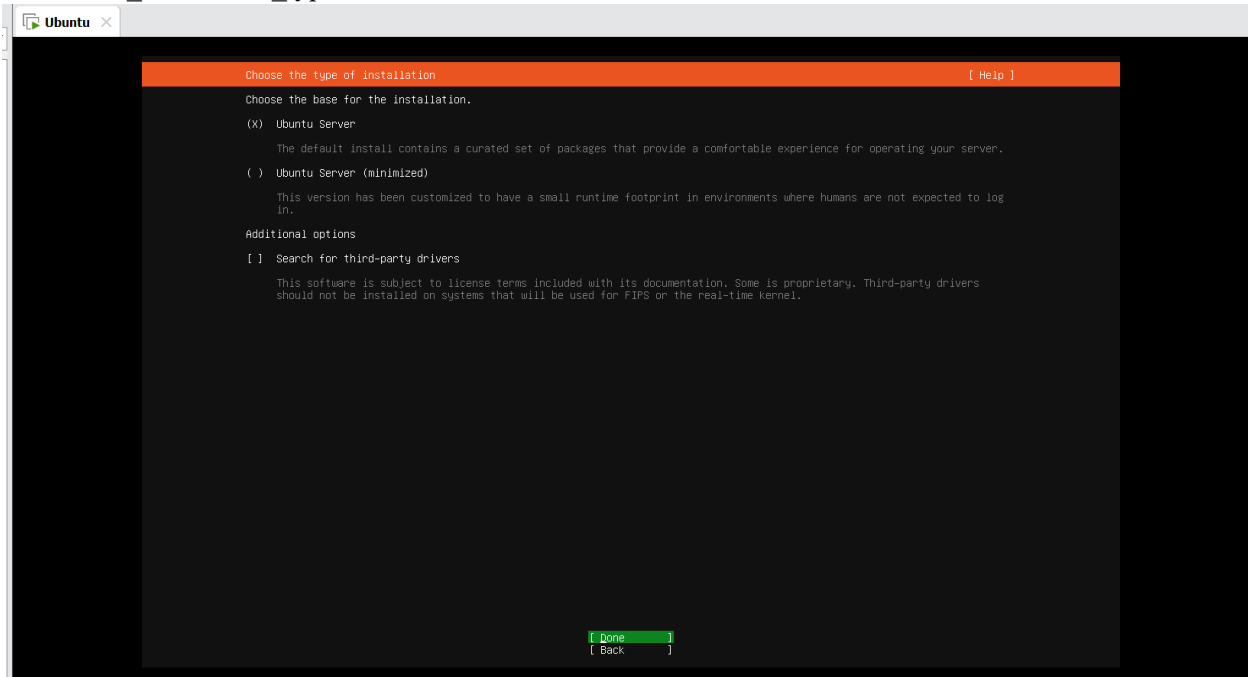
16.ubuntu\_keyboard\_variant



showing selected ubuntu keyboard variant.

## Task 7: Choose the Ubuntu Server Installation Type

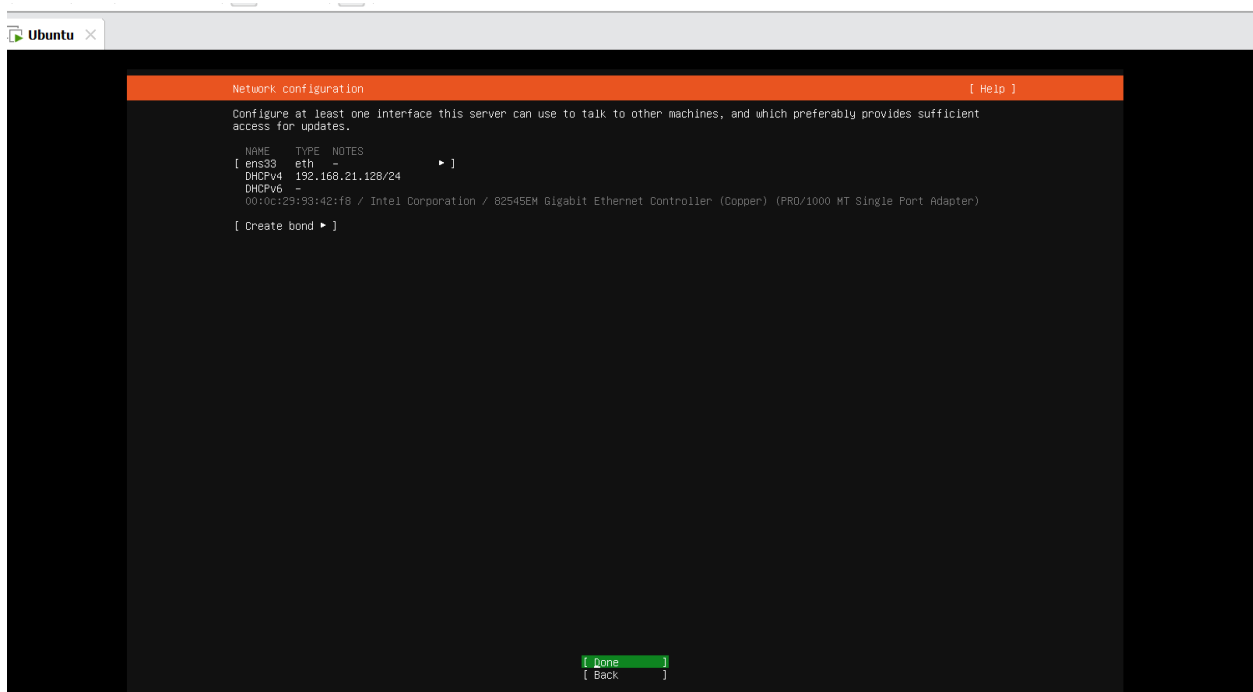
### 17. ubuntu\_installation\_type



Showing normal ubuntu server option selected.

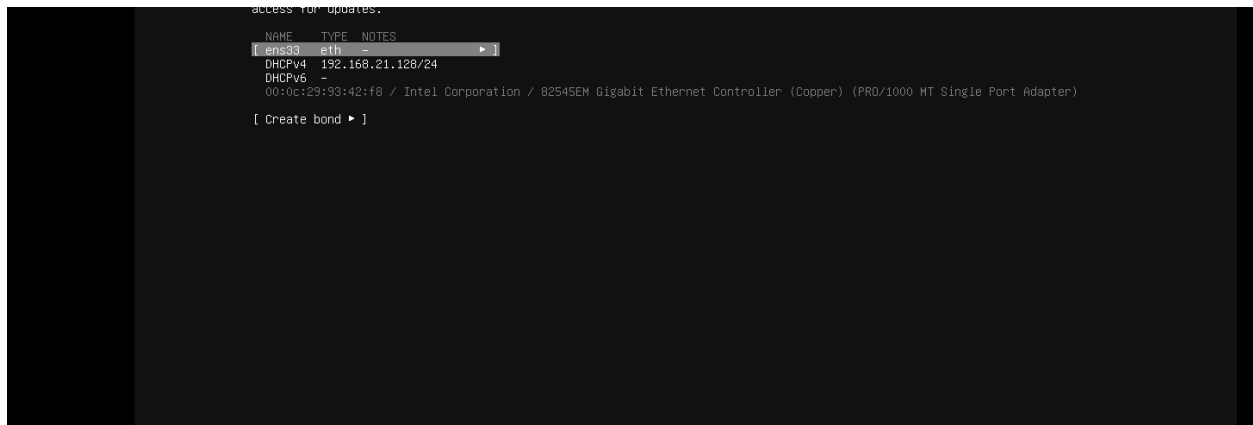
## Task 8: Configure the Network

### 18. ubuntu\_network\_interface



showing ubuntu network interface selected.

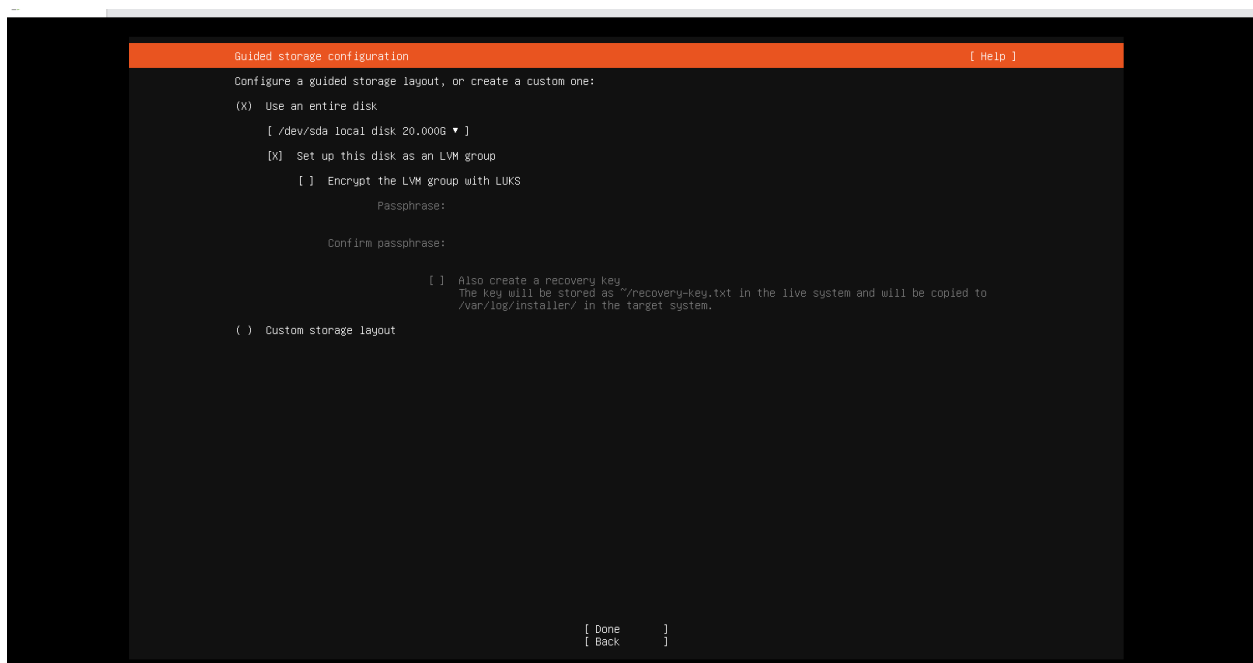
## 19. ubuntu\_installer\_network



Showing ubuntu server network interface showing ip address.

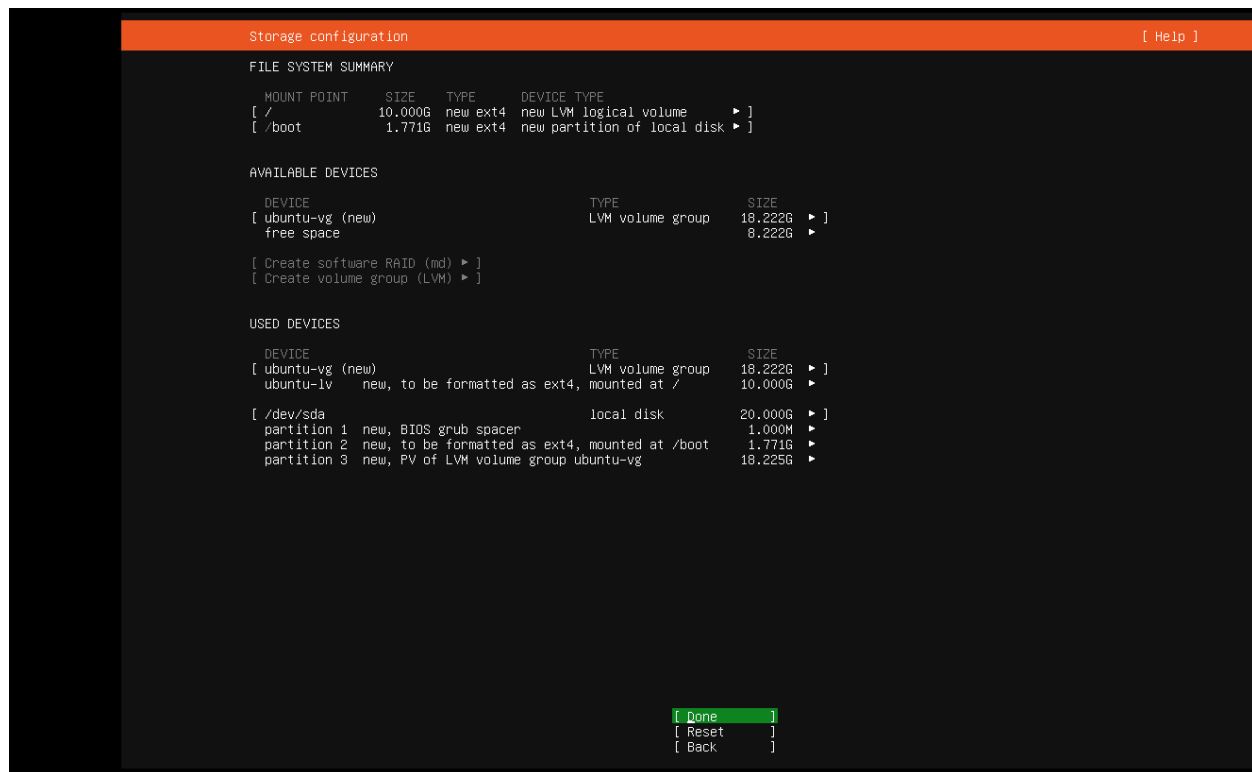
## Task 9: Configure Storage

## 20. ubuntu\_guided\_storage



Showing guided storage option selected.

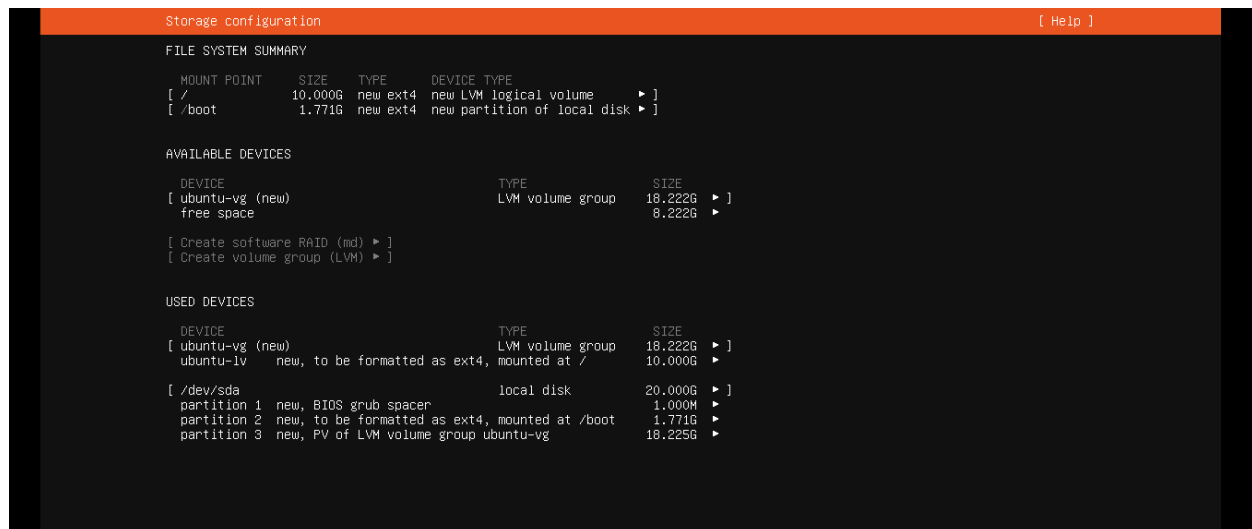
## 21. ubuntu\_storage\_configuration



Showing ubuntu server storage configuration.

## Task 10: Select the Installation Disk

### 22. ubuntu\_selected\_disk



showing the selected disk

### 23. ubuntu\_disk\_details

```
FILE SYSTEM SUMMARY

MOUNT POINT      SIZE      TYPE      DEVICE TYPE
[ /              10.000G   new ext4   new LVM logical volume ► ]
[ /boot          1.771G   new ext4   new partition of local disk ► ]

AVAILABLE DEVICES

DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                      LVM volume group                   18.222G ► ]
free space                               8.222G ► ]

[ Create software RAID (md) ► ]
[ Create volume group (LVM) ► ]

USED DEVICES

DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                      LVM volume group                   18.222G ► ]
ubuntu-lv    new, to be formatted as ext4, mounted at / 10.000G ► ]

[ /dev/sda                                local disk                          20.000G ► ]
partition 1  new, BIOS grub spacer                               1.000M ► ]
partition 2  new, to be formatted as ext4, mounted at /boot 1.771G ► ]
partition 3  new, PV of LVM volume group ubuntu-vg           18.225G ► ]
```

Showing ubuntu virtual disk name and size.

## Task 11: Review the Partitions

24. ubuntu\_partition\_layout

```
USED DEVICES

DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                      LVM volume group                   18.222G ► ]
ubuntu-lv    new, to be formatted as ext4, mounted at / 10.000G ► ]

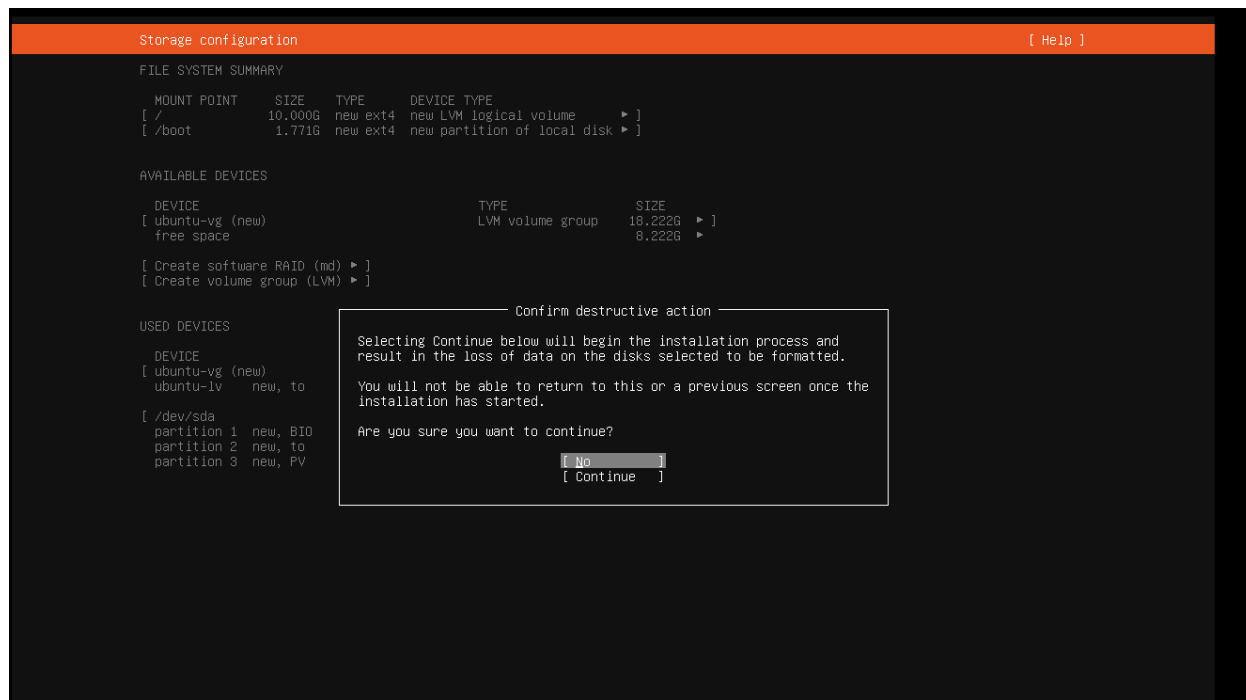
[ /dev/sda                                local disk                          20.000G ► ]
partition 1  new, BIOS grub spacer                               1.000M ► ]
partition 2  new, to be formatted as ext4, mounted at /boot 1.771G ► ]
partition 3  new, PV of LVM volume group ubuntu-vg           18.225G ► ]

[ Done ]
[ Reset ]
[ Back ]
```

Showing ubuntu partition layout.

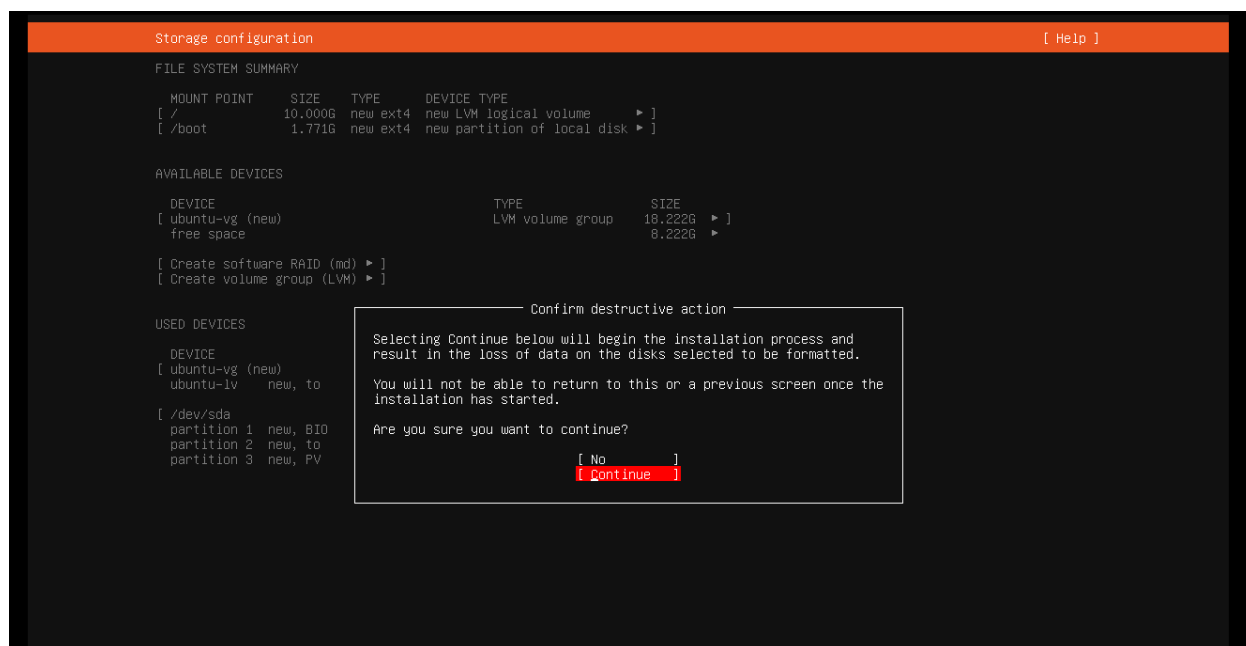
## Task 12: Confirm the Storage Changes

25. ubuntu\_storage\_warning



**Showing Final storage warning.**

**26. ubuntu\_storage\_confirmed**



**Showing storage confirmation**

**Task 13: Configure the Ubuntu Profile**

## 27. ubuntu\_actual\_name

:

Profile configuration

[ Help ]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Zahra Akram

Your servers name:   
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

### Showing ubuntu profile name.

## 28. ubuntu\_server\_name

Profile configuration

[ Help ]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Zahra Akram

Your servers name: ubuntu\_   
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

### Showing ubuntu profile server name

## 29. ubuntu\_username

Profile configuration

[ Help ]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Zahra Akram

Your servers name: ubuntu   
The name it uses when it talks to other computers.

Pick a username: zahraakram40

Choose a password:

### Showing ubuntu profile username.

## 30. ubuntu\_profile\_setup

Profile configuration [ Help ]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Zahra Akram

Your servers name: ubuntu  
The name it uses when it talks to other computers.

Pick a username: zahraakram40

Choose a password: \*\*\*\*\*

Confirm your password: \*\*\*\*\*

Showing complete ubuntu profile setup

## Task 14: Complete the Software Installation

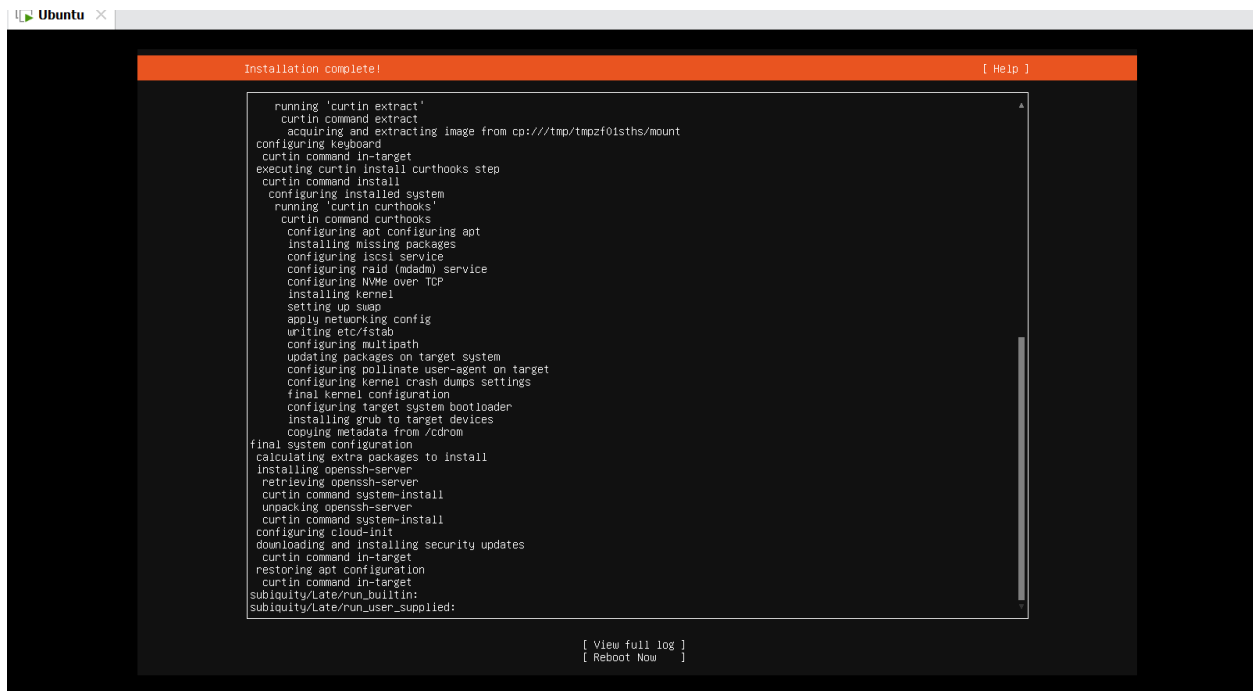
31. ubuntu\_installation\_progress

```
subiquity/load_cloud_config/extract_autoinstall:
subiquity/Early/apply_autoinstall_config:
subiquity/Reporting/apply_autoinstall_config:
subiquity/Error/apply_autoinstall_config:
subiquity/Userdata/apply_autoinstall_config:
subiquity/Package/apply_autoinstall_config:
subiquity/Debconf/apply_autoinstall_config:
subiquity/Kernel/apply_autoinstall_config:
subiquity/KernelCrashDumps/apply_autoinstall_config:
subiquity/Zdev/apply_autoinstall_config:
subiquity/Ad/apply_autoinstall_config:
subiquity/Late/apply_autoinstall_config:
configuring apt
curtin command in-target
installing system
executing curtin install initial step
executing curtin install partitioning step
curtin command install
configuring storage
running 'curtin block-meta simple'
curtin command block-meta
removing previous storage devices
configuring disk: disk-sda
configuring partition: partition-0
configuring partition: partition-1
configuring partitions: partition-2
configuring format: format-0
configuring lvm_voigroup: lvm_voigroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpnu5nzu52/mount |
```

Showing ubuntu installation progress

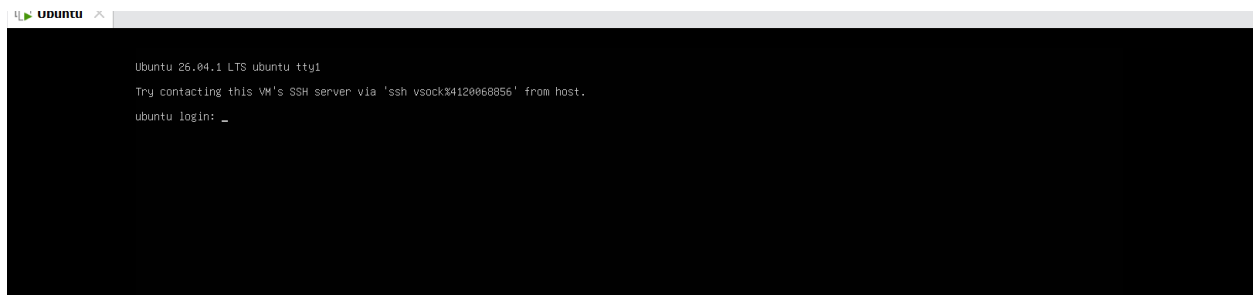
## Task 15: Reboot and verify the Ubuntu Identity

32. ubuntu\_installation\_complete



**Showing ubuntu installation completed.**

### 33. ubuntu\_login\_screen



**Showing ubuntu login screen**

### 34. ubuntu\_terminal\_login

```

ubuntu login: zahraakram40
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 04:35:08 PM UTC 2026

System load:  0.01          Processes:      217
Usage of /:   50.6% of 9.75GB Users logged in:  0
Memory usage: 21%          IPv4 address for ens33: 192.168.21.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

zahraakram40@ubuntu:~$

```

## Showing ubuntu terminal login

### 35. ubuntu\_identity\_verified

```

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

zahraakram40@ubuntu:~$

```

## Showing ubuntu identity verified by github username and password.

### Task 16: Find the Ubuntu Server IP Address

### 36. ubuntu\_ip\_addr\_command

```

zahraakram40@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:93:42:f8 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c299342f8
    inet 192.168.21.129/24 metric 100 brd 192.168.21.255 scope global dynamic ens33
        valid_lft 1772sec preferred_lft 1772sec
    inet6 fe80::20c:29ff:fe93:42f8/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
zahraakram40@ubuntu:~$ _

```

## Showing ubuntu ip address command.

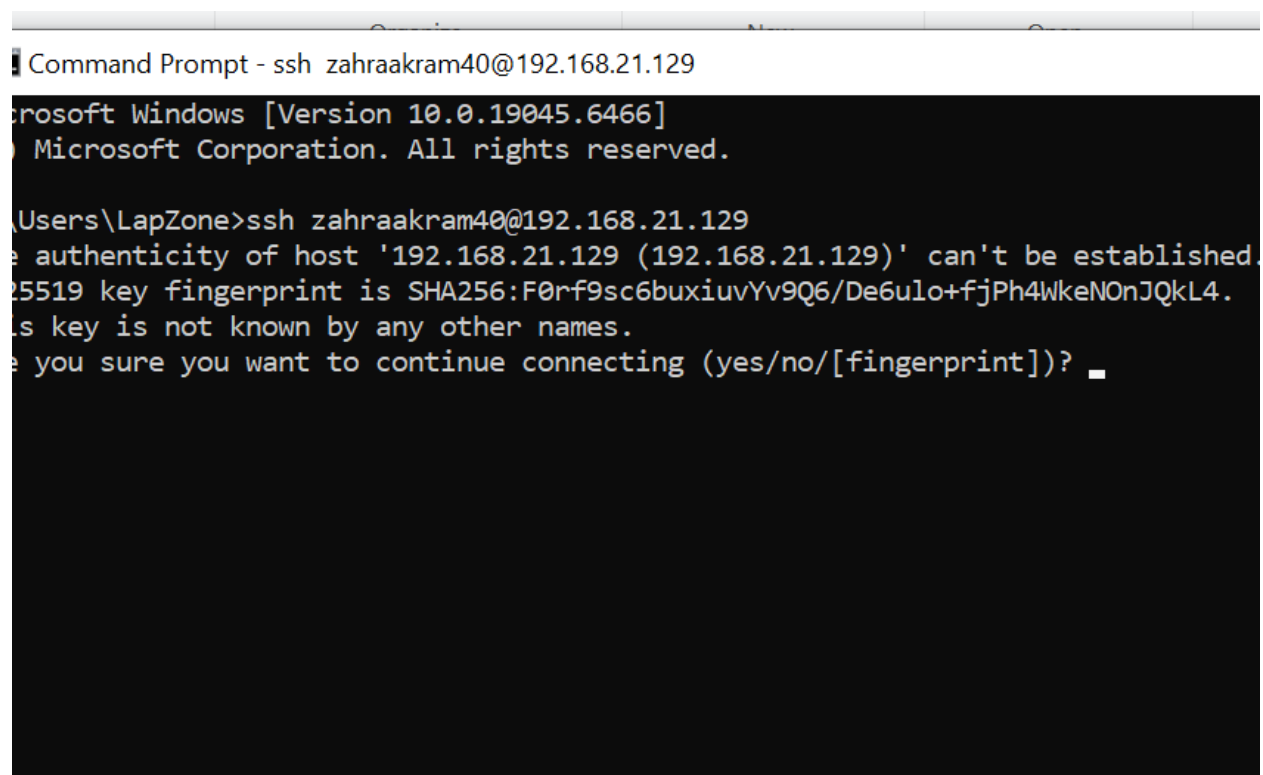
### 37. ubuntu\_ip\_address

```
link/ether 00:0c:29:93:42:f8 brd ff:ff:ff:ff:ff:ff
altname enp2s1
altnam enx000c299342f8
inet 192.168.21.129/24 metric 100 brd 192.168.21.255 scope global dynamic ens33
    valid_lft 1772sec preferred_lft 1772sec
inet6 fe80::20c:29ff:fe93:42f8/64 scope link proto kernel_l1
    valid_lft forever preferred_lft forever
zahraakram40@ubuntu:~$ _
```

**Showing ipv4 address.**

## **Task 17: Connect to Ubuntu Server from Windows**

38. windows\_ssh\_command



```
Command Prompt - ssh zahraakram40@192.168.21.129

Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\LapZone>ssh zahraakram40@192.168.21.129
Warning: The authenticity of host '192.168.21.129 (192.168.21.129)' can't be established.
ED25519 key fingerprint is SHA256:F0rf9sc6buxiuvYv9Q6/De6ulo+fjPh4WkeNOnJQkL4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? _
```

**Showing ssh command in command prompt.**

39. windows\_ssh\_fingerprint

```
Command Prompt - ssh zahraakram40@192.168.21.129
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\LapZone>ssh zahraakram40@192.168.21.129
The authenticity of host '192.168.21.129 (192.168.21.129)' can't be established.
ED25519 key fingerprint is SHA256:F0rf9sc6buxiuvYv9Q6/De6u1o+fjPh4WkeN0nJQkL4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.21.129' (ED25519) to the list of known hosts.
zahraakram40@192.168.21.129's password: 
```

**Showing accepted ssh server fingerprint.**

40. windows\_ssh\_login

```
zahraakram40@ubuntu: ~
System load:  0.02          Processes:           223
Usage of /:   47.7% of 9.75GB Users logged in:          0
Memory usage: 23%          IPv4 address for ens33: 192.168.21.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Sep 18 05:08:19 2026 from 192.168.21.1
zahraakram40@ubuntu:~$
```

**Showing successful ssh session from windows.**

41. windows\_ssh\_identity

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Sep 18 05:08:19 2026 from 192.168.21.1
zahraakram40@ubuntu:~$
```

**Showing username@ubuntu.**